



A Report on “ IDEA TO PRODUCT ”

A Complete Product Design Workshop

75
BIRLA VISHVAKARMA MAHAVIDYALAYA
VALLABH VIDYANAGAR
WORK IS WORSHIP

TRS STUDENT CHAPTER - BVM

idea to
PRODUCT

Between 21st August to 11th October '23
(9 Days)

C-206, Mechanical Engineering Dept

Scan the QR Code for info and Registration

TRS BVM
trs_bvm
TRS BVM Student Chapter
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Organised By

THE ROBOTICS SOCIETY STUDENT CHAPTER

B.V.M Engineering College [An Autonomous Institution]

Event Overview:

The Robotics Society (TRS) at Birla Vishvakarma Mahavidyalaya organized a 9-day, 18-hour workshop on "Idea to Product," aimed at empowering participants with a comprehensive understanding of developing a new product from ideation to realization. The workshop offered insights into Fusion 360 CAD design, 3D printing, Arduino IDE, and Tinkercad circuit simulation, providing hands-on experience to participants.

The first version of workshop conducted by Dr. Mehfuza Holia, Prof Jayesh Koisha, and Darshit Darji in 2022.

The second version of the workshop was coordinated by Dr. V.J. Patel, Dr. Milendrakumar Solanki, and Prof. Jayesh Koisha, with 24 out of 36 registered students completing it successfully.

Project Groups:

Promising features of this workshop:

- Hands-on workshop on 3D Design using Fusion 360
- Introduction to 3D printing with a demonstration
- Electronics component schematic design
- Introduction to Arduino IDE
- Simulation of the circuit in Tinkercad
- Hands-on soldering and assembly of electronics
- Final product assembly using 3D printed parts

Nine groups were formed, each tasked with designing a product such as a

- Laser proximity alarm,
- Digital clock
- Metal detector
- Touchless doorbell
- Touchless handwash

In the end, Four groups successfully transformed their ideas into products..

Workshop Topics:

The workshop covered Fusion 360 CAD design, 3D printing, Arduino IDE, and Tinkercad circuit simulation, enriching the students' learning experience.

Inaugural Ceremony:

The workshop began with an inaugural ceremony on August 21, 2023, attended by esteemed guests including Dr. Indrajit Patel, Principal BVM, Dr. Vinay Patel, Head Mechanical Dept and Advisor TRS student chapter, Dr. Tanmay Pawar, Head Electronics Department, Dr. Bhargav Goradia, Head EC dept, Dr. Milendrakumar Solanki and Prof. Jayesh Koisha , Faculty Event coordinators.

Amisha Sakhare, Kunal Engineer, Yash Sangale, Dev Bhanderi, Vivek Chauhan, Student coordinators. Total 36 enthusiastic students had the unique opportunity to engage in hands-on activities, thereby embarking on a journey of learning and innovation.

Objectives:

- Provide a comprehensive understanding of transforming ideas into products.
- Equip participants with skills for ideation, prototyping, and market launch.
- Offer insights into fundamental electronics components and simulation software.
- Introduce product design using Fusion 360 and 3D printing techniques.

Key Speaker:

Sessions were conducted by Dr. Milendrakumar Solanki, Prof. Jayesh Koisha from B.V.M. Engineering College and industry expert Mr. Ronak Dua from 3D-Curiosity Pvt. Ltd., sharing extensive expertise in 3D modeling..

Outcomes:

1. **Idea Generation:** Participants were encouraged to brainstorm and develop their ideas, with guidance from mentors and facilitators.
2. **Hardware Requirement:** Participants were learn how to choose the Electronics components as per the need of the product. How to simulate it on simulation software like tinkercad.
3. **Prototyping:** Participants learned about different prototyping techniques and tools to create a prototype of their product. They learnt how to make 3D model on software like Fusion 360 and 3D printing techniques to develop the different part of the Product.

The workshop concluded successfully, empowering students with practical skills and knowledge essential for product development.

Glimpses of an Events:



